Project 18 – Presentation Script (6 speakers, ~17 min)

Surface Water Quality Prediction · 14 slides · speaking pace ~130 words per minute

Strict Presentation Ground Rules for All 6 Speakers: Say all numerical metrics exactly as they appear on the slides. Never claim real-time sensor forecasting, hardware deployment, or laboratory replacement—our work is an offline surrogate on historical records. Emphasize "validation scores" on random hold-outs. Keep each verbal handover to exactly one crisp sentence.

Plan at a Glance: 6-Member Team Roster & Slide Allocation

Speaker	Role & Defense Zone	Slides	Target Time
Aman Soni (Lead)	Model Training Lead (Baseline, Random Forest, XGBoost)	1, 2, 9, 14	3:30
Pranav Sehgal	Research, Documentation & Presentation Lead (Literature, CCME WQI, TreeSHAP)	3, 12, 13	4:00
Mayank Sharma	EDA & Repository Management Lead (Dataset Scale, Sharding, Baseline Heatmaps)	4, 6	2:00
Aaryan Gera	Data Preprocessing Lead (Complete-Case Audit, Bounds Screening, 12 Pytest Tests)	5, 8	2:30
Chaitanya Kumar	Feature Engineering & Pipeline Lead (Stoichiometry, Skewness, Station Lags)	7	1:45
Ankit Kumar Pandey	Model Evaluation & Validation Lead (Metrics Benchmarking, Validity & Artifact Audit)	10, 11	3:00
Total	**Equitable 6-Member Team Presentation (Buffer for 12–20 min evaluation)**	**14 Slides**	**~17:00**

Slide 1 – Title · Aman Soni (Lead) (about 0:30)

Aman Soni · 0:30

> "Good morning everyone. We are Team 18, and our project is **Surface Water Quality Prediction**. We engineered models that estimate the Canadian CCME Water Quality Index from measured aquatic chemistry using Random Forest and XGBoost tree ensembles, supported by stoichiometric domain features.
>
> Alongside strong benchmark results, our paper and presentation present an honest, rigorous audit of what these scores actually prove and what they do not prove. Our team comprises Mayank Sharma, Aaryan Gera, Chaitanya Kumar, Ankit Kumar Pandey, Pranav Sehgal, and myself, Aman Soni. I will begin by defining the real-world operational problem."

Slide 2 – The Problem · Aman Soni (Lead) (about 1:00)

Aman Soni · 1:00

> "Today, freshwater monitoring relies on field grab samples sent back to centralized wet chemistry laboratories. Certain assays take days—most notably 5-day Biochemical Oxygen Demand, which requires a full 120-hour incubation period. Because composite water quality indices depend on these assays, river managers face an information delay of nearly a week before knowing if an aquatic ecosystem is in distress.
>
> Our objective is to train a fast mathematical surrogate: taking standard chemistry measurements—such as pH, temperature, dissolved oxygen, BOD, ammonia, orthophosphate, nitrogen, and nitrate—and mapping them instantly to the standardized CCME Water Quality Index score.
>
> To be scientifically precise about our scope of claims: we demonstrate offline regression on a historical 83-year multinational dataset. We do *not* claim real-time hardware deployment, field IoT sensing, laboratory replacement, or any neural network, which is planned for Phase 2. Now, I hand over to Pranav Sehgal to explain our mathematical target."

Slide 3 – Objective and Target (CCME WQI) · Pranav Sehgal (about 1:45)

Pranav Sehgal · 1:45

> "Thank you, Aman. The target variable our models learn is the Canadian Council of Ministers of the Environment Water Quality Index, or CCME WQI. It is formulated as: $CCME\ WQI = 100 - \sqrt{(F_1^2 + F_2^2 + F_3^2)} / 1.732$.
>
> Here, the factor 1.732—which is $\sqrt{3}$—scales a 3D Euclidean penalty vector to bound the final index cleanly between 0 and 100. The three sub-indices capture distinct ecological risks: **F_1 (Scope)** represents the percentage of variables failing guidelines; **F_2 (Frequency)** is the percentage of individual test failures; and **F_3 (Amplitude)** quantifies the relative magnitude of excursion via the normalized sum of excursions, or nse.
>
> The continuous index is categorized into five regulatory tiers: Poor (0 to 44, representing 4.5% of rows), Marginal (45 to 64, holding 11.6%), Fair (65 to 79, holding 11.2%), Good (80 to 94, holding 36.6%), and Excellent (95 to 100, holding 36.1%).
>
> Notice that our target distribution has an overall mean of approximately 85 and a median of 90.6—it is heavily left-skewed because over 72% of samples fall into Good or Excellent tiers. A crucial analytical property: F_1, F_2, and F_3 are not columns in the dataset; only the composite CCME_Values and tier labels are provided, both calculated from the same eight measurements we use as inputs. Predicting the index is therefore an exercise in non-linear function approximation. Now, Mayank Sharma will walk us through dataset scale."

Slide 4 – Dataset Scale and Composition · Mayank Sharma (about 1:00)

Mayank Sharma · 1:00

> "Thank you, Pranav. Our pipeline ingests **2,827,977 raw records** spanning 83 years of historical monitoring from 1940 to 2023 across five countries. After physical bound screening, we retain **2,819,617 clean observations** and engineer 38 model input features.
>
> However, an essential finding from our exploratory audit is extreme geographical imbalance: England's Environment Agency archive contributes 2,128,128 records, or **75.48%** of the clean data. The United States USGS provides 14.44%, Ireland's EPA provides 8.31%, China's Ministry of Ecology and Environment provides 1.63%, and Canada's ECCC provides 0.14% with 3,946 records.
>
> Furthermore, temporal coverage is heavily concentrated after the year 2000; pre-2000 historical records are sparse. Therefore, our model's baseline primarily reflects English river systems, and cannot support country-level generalizations without separate domain adaptation. Next, Aaryan Gera will present our data cleaning and bound screening."

Slide 5 – Data Audit and Physical-Bound Screening · Aaryan Gera (about 1:00)

Aaryan Gera · 1:00

> "Thank you, Mayank. When examining the delivered raw file, pandas reports zero null values across all 14 columns. However, our physical-bounds audit revealed **8,360 physically impossible sensor readings (0.296%)** that had to be purged.
>
> These violations included 1,581 records with pH outside [3.0, 12.0], 6,850 records with water temperature outside [0°C, 45°C], and corrupt negative concentrations: 2 negative ammonia, 2 negative BOD, and 10 negative orthophosphate readings. The individual checks sum to 8,445; their unique union is exactly 8,360, leaving 2,819,617 clean rows.
>
> We deliberately chose complete-case removal over mean or KNN imputation because these readings were corrupt sensor hardware dropouts—such as −999,999°C or pH 67,115. Imputing values into corrupted hardware records would invent synthetic data.
>
> We must also highlight a critical qualification: 'zero nulls' describes the delivered file. Several chemical columns cluster heavily at identical values—such as BOD with a median of 2.70 and 75th percentile of 2.83 mg/L. Moreover, repository audit graph 17 from an earlier uncurated version showed 26% to 34% missingness. This indicates upstream filling before final assembly is possible and unverified, so we do not claim this file represents uncurated authentic ground truth. Now back to Mayank for exploratory chemistry."

Slide 6 – Exploratory Findings · Mayank Sharma (about 1:00)

Mayank Sharma · 1:00

> "Thank you, Aaryan. On Slide 6, our bivariate scatter of 80,000 observations validates physical chemistry: dissolved oxygen follows Henry's law and the Benson–Krause saturation curve, dropping sharply as water temperature rises.
>
> When examining nutrient distributions, we observed extreme right-skewness: 12.0 for ammonia, 10.2 for BOD, and 9.0 for orthophosphate. Applying a `log(1+x)` transformation compresses these long tails and dramatically straightens linear correlation with CCME WQI: orthophosphate's Pearson correlation strengthens from −0.660 to −0.853; ammonia from −0.392 to −0.659; and BOD from −0.321 to −0.578.

> However, we present a crucial qualification noted at the bottom of the slide: *a higher Pearson r does not mean better prediction for trees*. Orthophosphate's non-parametric Spearman rank correlation is −0.917 for both raw and log(1+x). Because decision trees split purely on rank order, monotonic transforms leave tree splits completely unchanged. The log features primarily benefit linear models and future neural networks. Now, Chaitanya Kumar will present our domain feature engineering."

Slide 7 – Domain-Informed Feature Engineering · Chaitanya Kumar (about 1:45)

Chaitanya Kumar · 1:45

> "Thank you, Mayank. Rather than relying on black-box modeling, we engineered 30 domain-specific features across seven geochemical families, expanding our 8 base measurements to **38 model input features**:
>
> First, the **8 base chemical sensors**: pH, temperature, dissolved oxygen, BOD, ammonia, orthophosphate, total nitrogen, and nitrate.
> Second, **7 aquatic stoichiometry features**: Benson–Krause theoretical DO saturation percentage and deficit; Redfield-style N:P molar ratio; Emerson un-ionized toxic ammonia NH_3 calculated from pH-temperature dissociation; BOD-to-DO ratio; absolute neutral pH deviation; and nitrate proportion of nitrogen.
> Third, **7 log(1+x) variants** of skewed nutrient concentrations and ratios.
> Fourth, **6 station memory features**: shift-1 historical lags for DO, temperature, BOD, and orthophosphate, plus rolling-3 means for DO and temperature within each Area sorted by date, initializing the first record with the global median. Historical orthophosphate lag-1 achieves a strong Spearman correlation of −0.780 with the index.
> Fifth, **7 temporal features**: year, month, quarter, and cyclical sine/cosine encodings of month and day-of-year.
> Sixth, **2 categorical context features**: integer-encoded country and waterbody-type codes.
> Finally, **1 exceedance counter**: summing six regulatory guideline exceedances computed purely from input features.
>
> This 38-dimensional feature set encodes physical, stoichiometric, temporal, and spatial dynamics. Now, Aaryan Gera will explain pipeline architecture and testing."

Slide 8 – Pipeline Architecture and Testing · Aaryan Gera (about 1:30)

Aaryan Gera · 1:30

> "Thank you, Chaitanya. Slide 8 illustrates our end-to-end execution pipeline. We load 2.82 million cleaned rows with zero nulls, construct all 38 features, quarantine all target columns from the feature matrix, perform stratified 70/15/15 partitioning, train Random Forest and XGBoost with early stopping on validation, evaluate metrics, and extract built-in and TreeSHAP explainability.
>
> To ensure code reliability, we implemented an automated testing suite using pytest with **12 test functions** across four modules: features (4 tests), leakage (2 tests), metrics (3 tests), and models (3 tests), executing in 1.65 seconds with 100% pass rates.
>
> These unit tests explicitly verify physical bound screening, log1p monotonicity, shift-1 station lag isolation, metric mathematics, and assert that neither CCME_Values nor CCME_WQI can leak into input feature matrices.
>
> However, we must state an important architectural limitation: historical lags are built across the full dataset prior to partitioning. Because we use a random split, a station's previous record can sit in the training set while its current record sits in validation. While the unit tests prove target quarantine, they do not establish that predictions generalize to completely unseen monitoring stations or future years. Now, Aman Soni will detail our modeling strategy."

Slide 9 – Modeling Strategy and Evaluation Design · Aman Soni (Lead) (about 1:15)

Aman Soni · 1:15

> "Thank you, Aaryan. On Slide 9, we outline our modeling strategy. We partitioned our 2,819,617 cleaned rows using scikit-learn's `train_test_split`, stratified across the five WQI tiers with random seed 42: **1,973,731 training rows (70%)**, **422,943 validation rows (15%)**, and **422,943 test rows (15%)**.
>
> We trained three regressors directly on raw unscaled features: First, a **Mean Baseline Regressor** predicting the training set mean of 85.046, providing the performance floor any algorithm must beat. Second, a **Random Forest Regressor** configured with 30 estimators, maximum depth of 14, minimum samples per leaf of 5, and all 38 features evaluated at each split.
> Third, an **XGBoost Regressor** trained with histogram tree method, depth 6, learning rate 0.08, subsample 0.8, colsample 0.8, and early stopping after 20 rounds without validation improvement, converging at round 149.
>
> We clarify a discrepancy in earlier project drafts: earlier reports listed 100 trees at depth 20 for Random Forest and 300 rounds for XGBoost. Those did not match the actual saved code artifacts; the verified configurations here represent the true executed code.
>
> Finally, regarding deep learning: no neural network was implemented in this repository version. Multi-task deep learning is planned for Phase 2, and no premature results are claimed. Now, Ankit Kumar Pandey will present our validation benchmarking."

Slide 10 – Model Benchmarking · Ankit Kumar Pandey (about 1:30)

Ankit Kumar Pandey · 1:30

> "Thank you, Aman. Slide 10 presents our empirical validation benchmarks across 422,943 validation records:
>
> Our **Mean Baseline** yields a Mean Absolute Error of **14.130**, RMSE of **17.656**, and R^2 of **0.0000**.
> Our **Random Forest** slashes error to an MAE of **0.0521** index points, RMSE of **0.2081**, and R^2 of **0.99986**.
> For **XGBoost**, our comparison file records an MAE of **0.1375**, RMSE of **0.3740**, and R^2 of **0.99955**, while the training history log records MAE **0.1611**, RMSE **0.4372**, and R^2 **0.99939**.
>
> Notice the transparent dagger note on XGBoost: the stored predictions file reflects MAE 0.1375, while the saved model file records validation RMSE 0.4372, corresponding to MAE 0.1611. This indicates notebooks 03 and 04 should be re-run in sequence to align artifact provenance. Nevertheless, both runs confirm that tree ensembles reconstruct the CCME index within fractions of a single index point.
>
> Crucially, as highlighted in the bottom amber qualification card: these scores measure how well tree models reconstruct the mathematical CCME index from same-time chemical measurements on randomly held-out rows. They do *not* prove forecasting skill on future river conditions. Furthermore, inference latency was omitted because no timing harness or hardware logs exist in the repository, and the 422,943-row test set remains unscored. Now, I will present our reliability and validity audit on Slide 11."

Slide 11 – Reliability and Validity · Ankit Kumar Pandey (about 1:30)

Ankit Kumar Pandey · 1:30

> "On Slide 11, we address the critical scientific question: *Why does R^2 exceed 0.999, and what does the codebase actually demonstrate?* We highlight three open audit findings:
>
> **Finding 1: The Target is a Deterministic Formula of the Inputs.** CCME_Values is computed directly from the same eight chemical measurements that serve as model inputs, evaluated against static regulatory thresholds. High R^2 means our tree models successfully learned this deterministic non-linear equation—our repository's own evaluation notebook explicitly documents this function-approximation property.
> **Finding 2: exceedance_counter Re-Assessment.** In earlier audits, exceedance_counter was questioned. A code inspection confirms it is constructed strictly from six raw input measurements using independent thresholds; it does not contain target columns or F_1–F_3 values, so it is not a data-flow leak. However, it is mathematically redundant with the inputs and correlates at −0.777 with WQI. An ablation study dropping this feature is planned.
> **Finding 3: Random Split and Station History.** Because data partitioning was random rather than blocked by monitoring station or time, historical lags allow stations to share temporal context across partitions, and XGBoost used early stopping on validation.
>
> Therefore, we can honestly claim: **a fast, highly accurate computational surrogate of the CCME formula on this dataset**. We cannot claim generalization to unseen river basins or autonomous forecasting. Now, Pranav Sehgal will present model explainability."

Slide 12 – What Drives the Predictions · Pranav Sehgal (about 1:15)

Pranav Sehgal · 1:15

> "Thank you, Ankit. On Slide 12, we examine model explainability using built-in feature importance and TreeSHAP attribution. The dominant conclusion across both models is that **orthophosphate overwhelmingly drives predictions**.
>
> In Random Forest, raw orthophosphate accounts for 40.2% and its log(1+x) variant accounts for 48.3% of Gini impurity—combined, they represent **88.5%** of total forest importance. In XGBoost, raw orthophosphate contributes 64.0% and its log variant contributes 12.4% of gain—representing **76.4%** of total boosting gain. Ammonia and log ammonia follow at 5.1% and 4.4% in RF, and 6.9% and 7.8% in XGBoost.
>
> Our TreeSHAP analysis on 250 validation rows confirms this exact hierarchy: mean absolute SHAP impact is 8.27 index points for orthophosphate, 2.93 for log orthophosphate, and 2.39 for ammonia.
>
> However, we emphasize a vital qualification: *importance describes the model, not the river ecosystem*. Orthophosphate dominates because its stringent regulatory threshold of 0.05 mg/L sits right beside the dataset median of 0.040 mg/L, causing frequent large excursions that trigger F_1 and F_3 penalties. Additionally, because raw and log orthophosphate share identical ordering, the split between them is arbitrary. Station lag hist_ortho_lag1 accounts for 1.5% in XGBoost and is near zero in RF, refuting earlier claims that it ranked second. Now, I will conclude our limitations and Phase 2 roadmap on Slide 13."

Slide 13 – Limitations, Status and Next Steps · Pranav Sehgal (about 1:00)

Pranav Sehgal · 1:00

> "On Slide 13, we summarize our verified findings and project roadmap into three clear categories:
>
> **First, Confirmed from Code and Outputs:** Severe country imbalance (England represents 75.48% of records); random split without spatial or temporal hold-out; validation metrics only with test partition unscored; and a minor artifact discrepancy between comparison and training history files.
> **Second, Open and Unverified:** Investigating whether the zero-null file reflects upstream imputation due to value clustering and graph 17; executing an ablation experiment removing exceedance_counter; checking duplicate records; and establishing hardware latency benchmarks.
> **Third, Planned for Phase 2:** Implementing strict station-held-out (blocking by river basin) and time-forward splits (training pre-2015, testing post-2015); scoring the held-out test partition; re-running notebooks 03 and 04 to unify artifact provenance; and developing a multi-task deep neural network predicting both continuous scores and discrete tiers.
>
> We reiterate: no neural network code or metrics exist in this repository version, so none are claimed. Now, I hand back to our lead, Aman Soni, to conclude."

Slide 14 – Conclusion, Team and References · Aman Soni (Lead) (about 0:45)

Aman Soni · 0:45

> "Thank you, Pranav. In conclusion, Phase 1 delivers a clean, audited dataset of 2.82 million observations, 38 domain-engineered features, tested Random Forest and XGBoost benchmarks, and an interpretable SHAP attribution analysis. Tree ensembles successfully reconstruct the CCME Water Quality Index with validation MAE between 0.052 and 0.161 index points.
>
> Our team structure reflects equal, specialized contributions: Mayank Sharma led exploratory data analysis and repository management; Aaryan Gera led data preprocessing and pipeline testing; Chaitanya Kumar led stoichiometric feature engineering; Ankit Kumar Pandey led model evaluation and validity audits; Pranav Sehgal led research, documentation, and explainability; and I led model training.
>
> Our foundation rests on established literature: Breiman for Random Forests, Chen and Guestrin for XGBoost, Lundberg and Lee for SHAP, the CCME technical protocol, Benson and Krause for oxygen saturation, Redfield for nutrient stoichiometry, and Emerson for ammonia equilibrium. We are now ready for your questions. Thank you."

Emergency 12-Minute Fast-Track Trimming Guide

> **If the evaluation panel asks your team to finish within 12 minutes, execute these pre-planned cuts:**
>
> - **Slide 2 (Aman Soni):** Skip the grab sample step-by-step description; state the 120-hour BOD delay, project surrogate goal, and scope caveat. *(Saves 30s)*
> - **Slide 3 (Pranav Sehgal):** State the CCME WQI Euclidean formula and mention the 5 quality tiers; omit the granular tier percentage breakdown. *(Saves 45s)*
> - **Slide 6 (Mayank Sharma):** Highlight the Henry's law DO-temperature curve; state that log(1+x) straightens Pearson r without altering tree rank splits. *(Saves 25s)*

- **Slide 7 (Chaitanya Kumar):** Group the 38 features into 3 macro-groups (8 sensors, 14 chemistry/stoichiometry ratios, 16 temporal/spatial/context features). *(Saves 45s)*
- **Slide 8 (Aaryan Gera):** Mention the 12 pytest automated tests passing in 1.65s; skip individual module breakdowns. *(Saves 30s)*
- **Slide 10 (Ankit Kumar Pandey):** Report Baseline MAE 14.13, RF MAE 0.052, and XGBoost MAE 0.137–0.161; summarize artifact provenance in one sentence. *(Saves 30s)*
- **Slide 12 (Pranav Sehgal):** State that orthophosphate accounts for >76% gain in XGB and 88% in RF; state that SHAP confirms this ordering. *(Saves 30s)*
- **Slide 13 & 14 (Pranav & Aman):** Combine conclusion with Phase 2 next steps (station hold-outs and deep learning). *(Saves 45s)*

Total time reduced: ~5 minutes (Achieves a crisp, high-impact 12:00 presentation).

Viva Q&A Defense Master Guide (Role-Specific)

Q1: Why is your Random Forest R^2 equal to 0.9999? Is this extreme overfitting or data leakage?
Speaker: Aman Soni (Lead & Model Training Lead)
Answer: "It is neither data leakage nor typical overfitting. CCME WQI is an exact mathematical composite calculated deterministically from the same eight water chemistry measurements used as inputs. Because tree ensembles are universal non-linear approximators, training on nearly 2 million rows allows the trees to learn the underlying mathematical equation almost perfectly. In fact, Random Forest training MAE is 0.047 and validation MAE is 0.052—a 0.005 gap proving generalization across this dataset's random partition."

Q2: Why did you report two different XGBoost validation numbers (MAE 0.137 vs 0.161)?
Speaker: Ankit Kumar Pandey (Model Evaluation Lead)
Answer: "That reflects our rigorous artifact audit. The comparison table file (`model_comparison.csv`) recorded an MAE of 0.1375 and RMSE of 0.3740. However, when we audited the saved model object and its training history log, the recorded validation RMSE was 0.4372, corresponding to MAE 0.1611. This difference occurred because the stored predictions file came from a slightly different run than the serialized tree model. We transparently documented this dagger note in Table VI and Slide 10, confirming both show $R^2 > 0.999$."

Q3: Why did you not evaluate or report any deep learning neural networks?
Speaker: Aman Soni (Lead) or Pranav Sehgal (Research Lead)
Answer: "In Phase 1, our mandate was to establish an auditable, reproducible classical machine learning baseline. No neural network code or serialized weights exist in this repository version. Rather than making unverified claims, we explicitly state that deep learning (a multi-task MLP predicting continuous WQI and discrete tiers) is designated for Phase 2."

Q4: Why did you compute log(1+x) transforms if tree ensembles are invariant to monotonic transformations?
Speaker: Chaitanya Kumar (Feature Engineering Lead)
Answer: "Nutrients like orthophosphate and ammonia have skewness values exceeding 9.0 to 12.0. The log(1+x) transform was engineered to support linear baselines, correlation analysis, and Phase 2 deep learning neural networks, which are highly sensitive to extreme input scales. For tree models, rank order is preserved (Spearman rho remains −0.917), so tree split thresholds are functionally unchanged."

Q5: Does orthophosphate's 88.5% importance mean it is the primary biological cause of water degradation?
Speaker: Pranav Sehgal (Research & Explainability Lead)
Answer: "No. Feature importance describes model sensitivity, not interventional biological causality. Orthophosphate dominates because its stringent CCME guideline of 0.05 mg/L sits directly adjacent to the dataset median of 0.040 mg/L. As a result, small fluctuations trigger frequent and severe penalty excursions in the F_1 and F_3 sub-indices."

Q6: Why did you delete 8,360 rows instead of using mean, median, or KNN imputation?
Speaker: Aaryan Gera (Data Preprocessing Lead)
Answer: "Imputation is meant for missing data gaps. Our file contains zero null values. The 8,360 rows were physically impossible hardware sensor malfunctions—such as water temperature of −999,999°C and pH of 67,115. Imputing synthetic central tendencies over corrupt hardware dropouts would inject artificial bias into the data."

Q7: Why does your dataset have 75% of its records from England, and how does this affect validity?
Speaker: Mayank Sharma (EDA & Repository Management Lead)
Answer: "England's Environment Agency maintains one of the world's largest open-access automated surface water archives, contributing 2.13 million records. In contrast, Canada contributed 3,946 records. This geographic imbalance means our current model is primarily an English river surrogate. A model trained on English waterways cannot be assumed to transfer to tropical or arctic hydrological regimes without spatial domain adaptation."

Q8: Is exceedance_counter a data leakage feature?
Speaker: Chaitanya Kumar (Feature Engineering) or Ankit Kumar Pandey (Evaluation)
Answer: "No, it is not a data-flow leak. A code audit confirms exceedance_counter is computed exclusively from six raw chemical inputs (pH, DO, ammonia, BOD, nitrate, orthophosphate) using its own independent thresholds. It does not use target columns or F_1–F_3 terms. However, it is mathematically redundant with the inputs and strongly correlated with the target (−0.777), so an ablation without it is planned."